NEET UG CHEMISTRY — PHYSICAL CHEMISTRY

Thermodynamics

Comprehensive Exam-Oriented Tutorial, Worked Problems, and Original NEET UG Practice Assessment

> *"It is the only physical theory of universal content concerning which I am convinced that, within the framework of the applicability of its basic concepts, it will never be overthrown." — **Albert Einstein***

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1. Learning Objectives & Chapter Overview

After studying this tutorial, you will be able to:

- Explain the fundamental thermodynamic terms: **system**, **surroundings**, and **boundary**.
- Discriminate clearly between **open**, **closed**, and **isolated** systems.
- Explain **internal energy** (U), **work** (w), and **heat** (q) with IUPAC sign conventions.
- State the **First Law of Thermodynamics** and express it mathematically ($\Delta U = q + w$).
- Calculate energy changes as pressure-volume work and heat contributions in chemical and ideal gas systems (reversible, irreversible, isothermal, adiabatic, and free expansion).
- Explain state functions U and H (enthalpy), and correlate ΔU and ΔH via $\Delta H = \Delta U + \Delta n_g RT$.
- Describe experimental measurement of ΔU (bomb calorimeter) and ΔH (constant-pressure calorimeter).
- Define standard states and calculate enthalpy changes for phase transformations, formation, combustion, atomization, bond dissociation, lattice dissociation (Born-Haber cycle), solution, and dilution.
- State and apply **Hess's Law of Constant Heat Summation**.
- Differentiate between **extensive** and **intensive** properties, and derive $C_p - C_V = R$.
- Define spontaneous and non-spontaneous processes, and explain **entropy** (S) as a state function.
- State the **Second** and **Third Laws of Thermodynamics**.
- Explain **Gibbs energy change** (ΔG) and establish relationships between ΔG, reaction spontaneity, and the equilibrium constant (K).

Overview: Chemical energy stored by molecules can be released as heat during chemical reactions when a fuel like methane, cooking gas, or coal burns in air. It may also be used to do mechanical work in an engine or provide electrical energy through a galvanic cell like a dry cell. Thermodynamics deals with the study of these energy transformations in **macroscopic systems** involving a large number of molecules rather than microscopic systems containing a few molecules. Thermodynamics is not concerned with how and at what rate these transformations occur, but is based on the **initial and final equilibrium states** of a system. Macroscopic properties like pressure and temperature do not change with time for a system in an equilibrium state.

Through thermodynamics, we answer three central questions in chemistry:

1. *How do we determine the energy changes involved in a chemical reaction or process?*
2. *Will a given chemical reaction or process occur spontaneously or not, and what drives it?*
3. *To what extent do chemical reactions proceed before reaching equilibrium?*

2. Thermodynamic Terms & Classification of Systems

2.1 The System, Surroundings, and Boundary

In thermodynamics, a **system** refers to that part of the universe in which observations are made, and the remaining universe constitutes the **surroundings**. Together, the system and the surroundings constitute the universe:

$$\text{The Universe} = \text{The System} + \text{The Surroundings}$$

For all practical purposes, the surroundings are that portion of the remaining universe in the immediate neighbourhood of the system which can interact with it. For example, if we study a reaction between two substances A and B kept in a beaker, the reaction mixture inside the beaker is the system, and the room where the beaker is kept is the surroundings.

The system is separated from the surroundings by a real or imaginary wall called the **boundary**, which allows us to control and track all movements of matter and energy in or out of the system. A system may be defined by physical boundaries (like a beaker or test tube) or by a set of Cartesian coordinates specifying a particular volume in space.

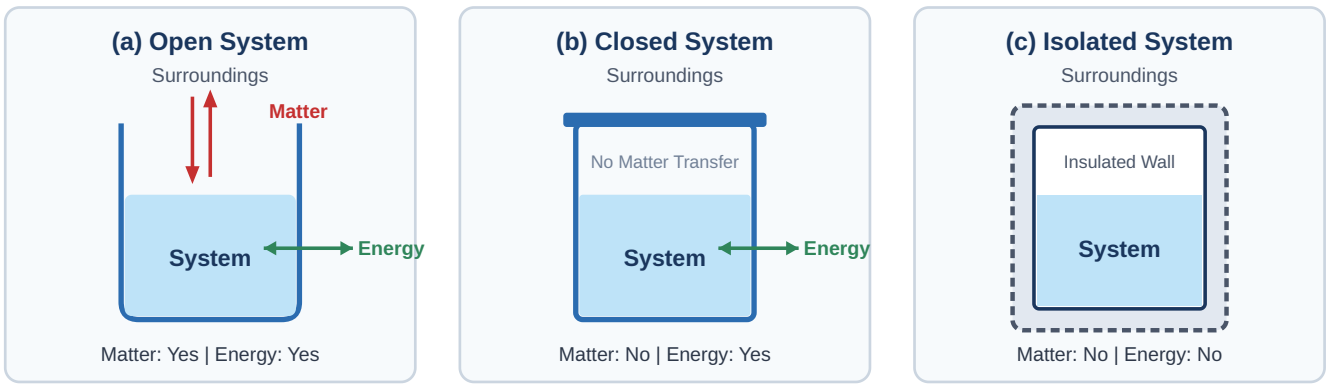
2.2 Types of Thermodynamic Systems

Systems are classified according to the movements of matter and energy across their boundary:

- **Open System:** Both **energy and matter** can be exchanged between the system and the surroundings. Example: Reactants present in an open beaker.
- **Closed System:** There is **no exchange of matter**, but **exchange of energy is possible** between the system and the surroundings. Example: Reactants in a closed vessel made of a thermally conducting material such as copper or steel.

BOSONANCE

- **Isolated System:** There is **no exchange of energy or matter** between the system and the surroundings. Example: Reactants kept in a closed thermos flask or any other closed, thermally insulated vessel.



Tutorial Diagram 1: Comparison of Open, Closed, and Isolated Thermodynamic Systems

2.3 The State of the System and State Functions

In mechanics, the state of a system at a given instant is specified by the position and velocity of each mass point. Thermodynamics introduces a simpler macroscopic concept: the **state of a thermodynamic system** is described by its measurable macroscopic (bulk) properties such as pressure (p), volume (V), temperature (T), and amount of substance (n).

- **State Functions (or State Variables):** Variables like p, V, T, U, H, S, G whose values depend **only on the current state of the system** and not on the path taken to reach that state. For example, changing the temperature of water from $25\ ^\circ\mathrm{C}$ to $35\ ^\circ\mathrm{C}$ gives $\Delta T = +10\ ^\circ\mathrm{C}$ regardless of whether we heat it directly or cool it first and then heat it. Similarly, the volume of water in a pond is independent of whether it was filled by rain, by a tubewell, or both.
- **Path Functions:** Quantities like heat (q) and work (w) whose values depend on the specific path or manner in which a change of state is carried out.

To completely define the state of a system, it is not necessary to define all properties, as only a minimum number of independent macroscopic properties need to be fixed; once fixed, all other properties automatically take definite values.

3. Internal Energy (U) and the First Law of Thermodynamics

3.1 Internal Energy as a State Function

The total energy of a system—representing the sum of chemical, electrical, mechanical, and all other forms of energy—is called its **internal energy (U)**. The internal energy of a system changes when:

1. Heat passes into or out of the system,
2. Work is done on or by the system, or
3. Matter enters or leaves the system.

Important Note on Absolute Values: Unlike volume, which has an unambiguous absolute value in a particular state, we cannot specify the absolute value of internal energy U. Thermodynamics measures only the **change in internal energy (ΔU)** accompanying a process.

3.2 Adiabatic Work (w_{ad}) and Joule's Experiments

An **adiabatic system** is thermally insulated so that no heat can be exchanged between the system and surroundings ($q = 0$) across its boundary (called an **adiabatic wall**). Any process occurring in such a system is an **adiabatic process**.

Between 1840 and 1850, J. P. Joule performed experiments on water kept in an adiabatic vessel initially at state A (temperature T_{A}, internal energy U_{A}):

- **Path 1 (Mechanical Work):** Doing $1\ \mathrm{kJ}$ of mechanical work by rotating a set of small paddles (churning water) raised the temperature to T_{B} (state B, internal energy U_{B}), giving $\Delta T = T_{\mathrm{B}} - T_{\mathrm{A}}$.
- **Path 2 (Electrical Work):** Doing an equal amount ($1\ \mathrm{kJ}$) of electrical work with an immersion rod produced the exact same temperature change $\Delta T = T_{\mathrm{B}} - T_{\mathrm{A}}$.

Joule proved that a given amount of adiabatic work done on a system produces the same change of state irrespective of the path, establishing that internal energy U is a **state function**:

$$\Delta U = U_2 - U_1 = w_{\mathrm{ad}}$$

3.3 Heat (q) and IUPAC Sign Conventions

Internal energy can also be changed by transferring heat across **thermally conducting walls** (such as a copper container placed in a heat reservoir) without doing work. At constant volume (no work done), $\Delta U = q$.

BOSONANCE

IUPAC Sign Conventions in Chemical Thermodynamics

- **Heat absorbed by the system** from the surroundings: $q > 0$ (positive, internal energy increases).
- **Heat released by the system** to the surroundings: $q < 0$ (negative, internal energy decreases).
- **Work done ON the system** (compression): $w > 0$ (positive, internal energy increases).
- **Work done BY the system** (expansion): $w < 0$ (negative, internal energy decreases).

3.4 Mathematical Statement of the First Law of Thermodynamics

When a change of state is brought about both by doing work and by transfer of heat, the change in internal energy is:

$$\Delta U = q + w$$

While q and w individually depend on the path, their sum $q + w = \Delta U$ depends only on the initial and final states. For an **isolated system**, $q = 0$ and $w = 0$, so $\Delta U = 0$.

First Law of Thermodynamics: *The energy of an isolated system is constant.* Equivalently, it is the law of conservation of energy: energy can neither be created nor destroyed.

4. Pressure-Volume Work & Thermodynamic Processes

4.1 Mechanical Work of Compression and Expansion

Consider a cylinder fitted with a frictionless piston of cross-sectional area A, containing $1\ \text{mol}$ of an ideal gas of initial volume V_i and internal pressure p. If a constant external pressure $p_{\text{ex}} > p$ pushes the piston inward a distance l in a single step until the final volume is V_f:

- Volume change: $\Delta V = V_f - V_i = -l \cdot A$
- Force on piston: $\text{Force} = p_{\text{ex}} \cdot A$
- Work done on the system: $w = \text{Force} \times \text{distance} = (p_{\text{ex}} \cdot A) \cdot l = -p_{\text{ex}}(V_f - V_i) = -p_{\text{ex}}\Delta V$

$$w = -p_{\text{ex}}\Delta V = -p_{\text{ex}}(V_f - V_i)$$

During compression, $V_f < V_i$, so ΔV is negative, and $w = -p_{\text{ex}}\Delta V$ becomes **positive** (work done on the system). If the external pressure changes in a series of finite steps, total work is $w = -\sum p_{\text{ex}}\Delta V$.

4.2 Reversible vs. Irreversible Processes

A process is said to be **reversible** if a change is brought about in such a way that the process could, at any moment, be reversed by an infinitesimal change. A reversible process proceeds infinitely slowly through a series of equilibrium states such that the system and surroundings are always in near-equilibrium with each other ($p_{\text{ex}} = p_{\text{in}} \pm dp$). Processes other than reversible processes are called **irreversible processes**.

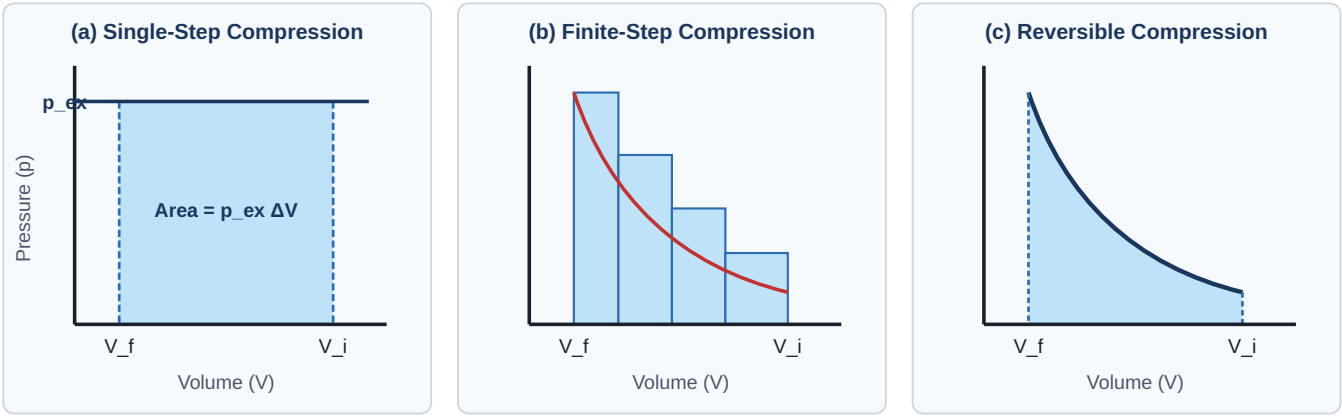
Under reversible conditions, since $dp \cdot dV$ is negligibly small:

$$w_{\text{rev}} = -\int_{V_i}^{V_f} p_{\text{ex}}\, dV = -\int_{V_i}^{V_f} (p_{\text{in}} \pm dp)\, dV = -\int_{V_i}^{V_f} p_{\text{in}}\, dV$$

For n moles of an ideal gas ($p = \dfrac{nRT}{V}$) undergoing an **isothermal reversible process** ($T = \text{constant}$):

$$w_{\text{rev}} = -\int_{V_i}^{V_f} nRT\,\frac{dV}{V} = -nRT\ln\left(\frac{V_f}{V_i}\right) = -2.303\,nRT\log_{10}\left(\frac{V_f}{V_i}\right)$$

BOSONANCE



Tutorial Diagram 2: Pressure-Volume (pV) plots showing work done on a gas (shaded area) during (a) single-step irreversible compression, (b) finite-step compression, and (c) reversible compression from initial volume V_i to final volume V_f.

4.3 Summary of Special Thermodynamic Processes for an Ideal Gas

- **Constant Volume (Isochoric Process, $\Delta V = 0$):** Since $w = -p_{\text{ex}}\Delta V = 0$, the First Law gives:
$$\Delta U = q_V$$
where q_V is the heat supplied at constant volume.
- **Free Expansion (Expansion into Vacuum, $p_{\text{ex}} = 0$):** No work is done during free expansion ($w = 0$) whether the process is reversible or irreversible. For an **isothermal free expansion** of an ideal gas, Joule determined experimentally that $q = 0$ and $\Delta U = 0$.
- **Isothermal Process ($T = \text{constant}$, so $\Delta U = 0$ for an ideal gas):**
  - Irreversible isothermal change: $q = -w = p_{\text{ex}}(V_f - V_i)$
  - Reversible isothermal change: $q = -w = nRT \ln\left(\frac{V_f}{V_i}\right) = 2.303\, nRT \log_{10}\left(\frac{V_f}{V_i}\right)$
- **Adiabatic Process ($q = 0$):** $\Delta U = w_{\text{ad}}$.

5. Enthalpy (H), Heat Capacity, and Calorimetry

5.1 Enthalpy (H) — A Useful State Function

Most laboratory chemical reactions are carried out in flasks or test tubes open to constant atmospheric pressure rather than at constant volume. At constant pressure, the First Law can be written as:

$$\Delta U = U_2 - U_1 = q_p - p(V_2 - V_1) \implies q_p = (U_2 + pV_2) - (U_1 + pV_1)$$

We define a new thermodynamic state function called **enthalpy (H)** (from the Greek *enthalpien*, to warm or heat content) as:

$$H = U + pV$$

Because U, p, and V are all state functions, H is also a **state function**. For finite changes at constant pressure ($\Delta p = 0$):

$$\Delta H = q_p = \Delta U + p\Delta V$$

- For **exothermic reactions** (which evolve heat to the surroundings), ΔH is **negative** ($\Delta H < 0$).
- For **endothermic reactions** (which absorb heat from the surroundings), ΔH is **positive** ($\Delta H > 0$).

5.2 Relationship Between ΔH and ΔU for Gaseous Reactions

For reactions involving only solids and liquids, volume changes upon heating or reacting are negligible ($\Delta H \approx \Delta U$). When gases are involved at constant temperature T and pressure p, let n_A (total volume V_A) be the moles of gaseous reactants and n_B (total volume V_B) be the moles of gaseous products. By the ideal gas law:

$$pV_A = n_A RT \quad \text{and} \quad pV_B = n_B RT \implies p(V_B - V_A) = p\Delta V = (n_B - n_A)RT = \Delta n_g RT$$

$$\Delta H = \Delta U + \Delta n_g RT$$

where $\Delta n_g = (\text{moles of gaseous products}) - (\text{moles of gaseous reactants})$.

5.3 Extensive and Intensive Properties

- **Extensive Properties:** Properties whose value **depends on the quantity or size of matter** present in the system. Examples: mass (m), volume (V), internal energy (U), enthalpy (H), entropy (S), Gibbs energy (G), heat capacity (C).
- **Intensive Properties:** Properties whose value **does not depend on the quantity or size of matter** present in the system. Examples: temperature (T), pressure (p), density (ρ), refractive index, viscosity, surface tension, and any **molar property** ($\chi_{\text{m}} = \chi/n$) such as molar volume (V_{m}), molar heat capacity (C_{m}), and specific heat capacity (c).

BOSONANCE

Partition Test: If a container of gas of volume V at temperature T is divided into two equal halves by a partition, each compartment has volume $\frac{V}{2}$ (showing V is extensive), while the temperature in each remains T (showing T is intensive).

5.4 Heat Capacity and the Relation $C_p - C_V = R$

The heat (q) absorbed by a system is proportional to its temperature rise (ΔT):

$$q = C\Delta T = nC_{\mathrm{m}}\Delta T = mc\Delta T$$

- **Heat Capacity (C):** Heat required to raise the temperature of the entire sample by $1\ ^{\circ}\mathrm{C}$ (or $1\ \mathrm{K}$). (Extensive property, unit: $\mathrm{J\ K^{-1}}$).
- **Molar Heat Capacity ($C_{\mathrm{m}} = C/n$):** Heat required to raise the temperature of $1\ \mathrm{mol}$ of a substance by $1\ ^{\circ}\mathrm{C}$ (or $1\ \mathrm{K}$). (Intensive property, unit: $\mathrm{J\ mol^{-1}\ K^{-1}}$).
- **Specific Heat Capacity ($c = C/m$):** Heat required to raise the temperature of unit mass ($1\ \mathrm{g}$ or $1\ \mathrm{kg}$) of a substance by $1\ ^{\circ}\mathrm{C}$ (or $1\ \mathrm{K}$). (Intensive property, unit: $\mathrm{J\ g^{-1}\ K^{-1}}$).

Derivation of $C_p - C_V = R$ for $1\ \mathrm{mol}$ of an Ideal Gas:
At constant volume, $q_V = C_V\Delta T = \Delta U$. At constant pressure, $q_p = C_p\Delta T = \Delta H$.
For $1\ \mathrm{mol}$ of an ideal gas, $\Delta H = \Delta U + \Delta(pV) = \Delta U + \Delta(RT) = \Delta U + R\Delta T$.
Substituting $\Delta H = C_p\Delta T$ and $\Delta U = C_V\Delta T$:

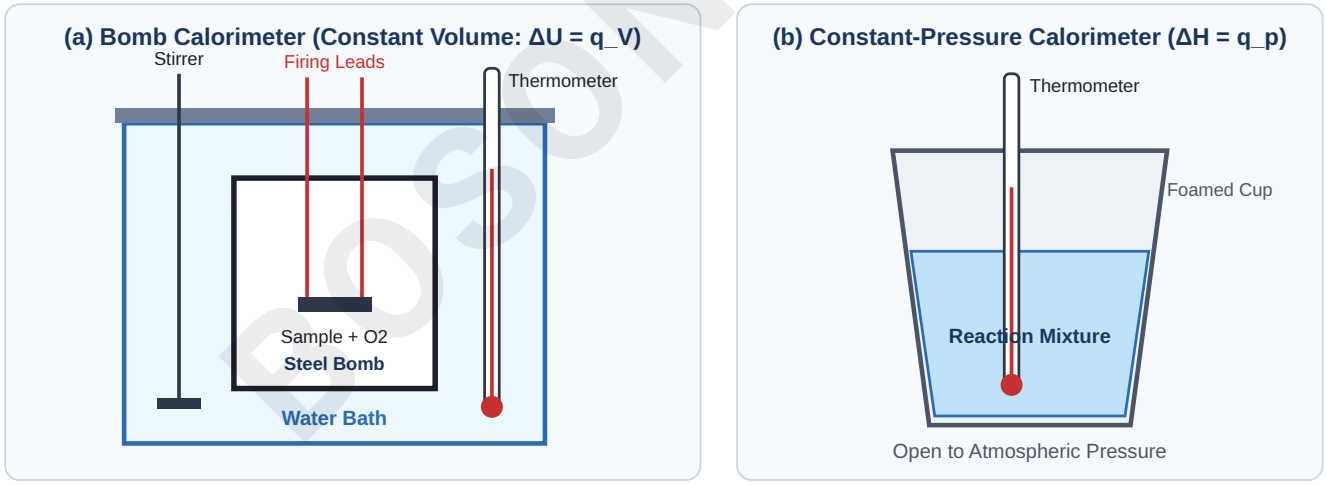
$$C_p\Delta T = C_V\Delta T + R\Delta T \implies C_p - C_V = R$$

5.5 Experimental Measurement of ΔU and ΔH: Calorimetry

1. Measurement of ΔU (Bomb Calorimeter): A combustible substance is burnt in pure dioxygen inside a sealed steel vessel (the bomb) immersed in a water bath at constant volume ($\Delta V = 0$). Because the volume does not change, no pressure-volume work is done ($w = 0$) even if gases are involved. The heat evolved transfers to the calorimeter and water bath, and the temperature rise ΔT is converted to $q_V = \Delta U$ using the heat capacity C_V of the calorimeter:

$$\Delta U = q_V = -C_V \times \Delta T$$

2. Measurement of ΔH (Constant-Pressure Calorimeter): Carried out in a foamed polystyrene cup open to constant atmospheric pressure. The heat change measured is $q_p = \Delta_r H$.



Tutorial Diagram 3: Experimental Calorimetry Apparatus for (a) Constant-Volume (ΔU) and (b) Constant-Pressure (ΔH) Measurements.

6. Reaction Enthalpy ($\Delta_r H$), Hess's Law, and Types of Enthalpies

6.1 Reaction Enthalpy and Standard States

The enthalpy change accompanying a chemical reaction is called the **reaction enthalpy ($\Delta_r H$)**:

$$\Delta_r H = \sum_i a_i H_{\text{products}} - \sum_i b_i H_{\text{reactants}}$$

where a_i and b_i are the stoichiometric coefficients of products and reactants in the balanced chemical equation.

Standard State ($\ominus$): The standard state of a substance at a specified temperature is its **pure form at a pressure of $1\ \mathrm{bar}$** (data are conventionally reported at $298\ \mathrm{K}$). For example, the standard state of liquid ethanol at $298\ \mathrm{K}$ is pure liquid ethanol at $1\ \mathrm{bar}$; the standard state of solid iron at $500\ \mathrm{K}$ is pure solid iron at $1\ \mathrm{bar}$. Standard enthalpy changes are denoted by $\Delta_r H^{\ominus}$.

BOSONANCE

6.2 Enthalpy Changes During Phase Transformations

- **Standard Enthalpy of Fusion ($\Delta_{\text{fus}}H^{\ominus}$):** The enthalpy change accompanying the melting of $1\ \text{mol}$ of a solid substance in its standard state at constant temperature (melting point T_f) and standard pressure ($1\ \text{bar}$). Melting is always endothermic ($\Delta_{\text{fus}}H^{\ominus} > 0$). Example:
$$\text{H}_2\text{O(s)} \rightarrow \text{H}_2\text{O(l)}; \quad \Delta_{\text{fus}}H^{\ominus} = +6.00\,\text{kJ mol}^{-1}$$
Freezing is the reverse process and releases an equal amount of heat ($\Delta_{\text{freeze}}H^{\ominus} = -\Delta_{\text{fus}}H^{\ominus}$).
- **Standard Enthalpy of Vaporization ($\Delta_{\text{vap}}H^{\ominus}$):** Amount of heat required to vaporize $1\ \text{mol}$ of a liquid at constant temperature (boiling point T_b) and under standard pressure ($1\ \text{bar}$). Example:
$$\text{H}_2\text{O(l)} \rightarrow \text{H}_2\text{O(g)}; \quad \Delta_{\text{vap}}H^{\ominus} = +40.79\,\text{kJ mol}^{-1}$$
- **Standard Enthalpy of Sublimation ($\Delta_{\text{sub}}H^{\ominus}$):** The change in enthalpy when $1\ \text{mol}$ of a solid substance converts directly into vapour at a constant temperature and under standard pressure ($1\ \text{bar}$). Example: Solid CO_2 ('dry ice') sublimes at $195\ \text{K}$ with $\Delta_{\text{sub}}H^{\ominus} = 25.2\,\text{kJ mol}^{-1}$; naphthalene has $\Delta_{\text{sub}}H^{\ominus} = 73.0\,\text{kJ mol}^{-1}$.

The magnitude of phase transformation enthalpies depends on the strength of intermolecular forces (e.g., strong hydrogen bonding in water vs. weaker dipole-dipole interactions in acetone).

Substance	Melting Point T_f/K	$\Delta_{\text{fus}}H^{\ominus}/(\text{kJ mol}^{-1})$	Boiling Point T_b/K	$\Delta_{\text{vap}}H^{\ominus}/(\text{kJ mol}^{-1})$
N_2	63.15	0.72	77.35	5.59
NH_3	195.40	5.65	239.73	23.35
HCl	159.0	1.992	188.0	16.15
CO	68.0	6.836	82.0	6.04
CH_3COCH_3 (Acetone)	177.8	5.72	329.4	29.1
CCl_4	250.16	2.5	349.69	30.0
H_2O	273.15	6.01	373.15	40.79
NaCl	1081.0	28.8	1665.0	170.0
C_6H_6 (Benzene)	278.65	9.83	353.25	30.8

6.3 Standard Molar Enthalpy of Formation ($\Delta_f H^{\ominus}$)

The **standard molar enthalpy of formation ($\Delta_f H^{\ominus}$)** is the standard enthalpy change for the formation of **one mole of a compound** from its constituent elements in their most stable states of aggregation (called **reference states**) at $25\ ^{\circ}\text{C}$ ($298\ \text{K}$) and $1\ \text{bar}$ pressure.

- Reference states of common elements: dihydrogen is $\text{H}_2(\text{g})$, dioxygen is $\text{O}_2(\text{g})$, carbon is $\text{C}(\text{graphite, s})$, sulphur is $\text{S}(\text{rhombic, s})$, and bromine is $\text{Br}_2(\text{l})$.
- **Convention:** The standard enthalpy of formation ($\Delta_f H^{\ominus}$) of any element in its reference state is taken as **zero**.
- Using $\Delta_f H^{\ominus}$ values, the standard enthalpy of any reaction is calculated as:

$$\Delta_r H^{\ominus} = \sum_i a_i \Delta_f H^{\ominus}(\text{products}) - \sum_i b_i \Delta_f H^{\ominus}(\text{reactants})$$

Crucial Distinction: When is $\Delta_r H^{\ominus}$ Equal to $\Delta_f H^{\ominus}$?

A reaction represents $\Delta_f H^{\ominus}$ **only** when **exact** $1\ \text{mol}$ of the compound is formed directly from its constituent elements in their reference states:

- $\text{CaO(s)} + \text{CO}_2\text{(g)} \rightarrow \text{CaCO}_3\text{(s)}$ ($\Delta_r H^{\ominus} = -178.3\,\text{kJ mol}^{-1}$) is **NOT** $\Delta_f H^{\ominus}(\text{CaCO}_3)$ because it is formed from compounds, not elements.
- $\text{H}_2\text{(g)} + \text{Br}_2\text{(l)} \rightarrow 2\text{HBr(g)}$ ($\Delta_r H^{\ominus} = -72.8\,\text{kJ mol}^{-1}$) forms $2\ \text{mol}$ of HBr, so $\Delta_r H^{\ominus} = 2\Delta_f H^{\ominus}(\text{HBr})$. Thus $\Delta_f H^{\ominus}[\text{HBr(g)}] = -36.4\,\text{kJ mol}^{-1}$.

BOSONANCE

Substance	$\Delta_f H^\ominus / (\mathrm{kJ\,mol^{-1}})$	Substance	$\Delta_f H^\ominus / (\mathrm{kJ\,mol^{-1}})$
$Al_2O_3(s)$	-1675.7	$HI(g)$	$+26.48$
$BaCO_3(s)$	-1216.3	$KCl(s)$	-436.75
$Br_2(l)$	0	$KBr(s)$	-393.8
$Br_2(g)$	$+30.91$	$MgO(s)$	-601.70
$CaCO_3(s)$	-1206.92	$Mg(OH)_2(s)$	-924.54
$C(\text{diamond})$	$+1.89$	$NaF(s)$	-573.65
$C(\text{graphite})$	0	$NaCl(s)$	-411.15
$CaO(s)$	-635.09	$NaBr(s)$	-361.06
$CH_4(g)$	-74.81	$NaI(s)$	-287.78
$C_2H_4(g)$	$+52.26$	$NH_3(g)$	-46.11
$CH_3OH(l)$	-238.86	$NO(g)$	$+90.25$
$C_2H_5OH(l)$	-277.69	$NO_2(g)$	$+33.18$
$C_6H_6(l)$	$+49.0$	$PCl_3(l)$	-319.70
$CO(g)$	-110.53	$PCl_5(s)$	-443.5
$CO_2(g)$	-393.51	$SiO_2(s, \text{quartz})$	-910.94
$C_2H_6(g)$	-84.68	$SnCl_2(s)$	-325.1
$Cl_2(g)$	0	$SnCl_4(l)$	-511.3
$C_3H_8(g)$	-103.85	$SO_2(g)$	-296.83
$\text{n-}C_4H_{10}(g)$	-126.15	$SO_3(g)$	-395.72
$HgS(s, \text{red})$	-58.2	$SiH_4(g)$	$+34$
$H_2(g)$	0	$SiCl_4(g)$	-657.0
$H_2O(g)$	-241.82	$C(g)$	$+716.68$
$H_2O(l)$	-285.83	$H(g)$	$+217.97$
$HF(g)$	-271.1	$Cl(g)$	$+121.68$
$HCl(g)$	-92.31	$Fe_2O_3(s)$	-824.2
$HBr(g)$	-36.40	–	–

6.4 Conventions for Thermochemical Equations

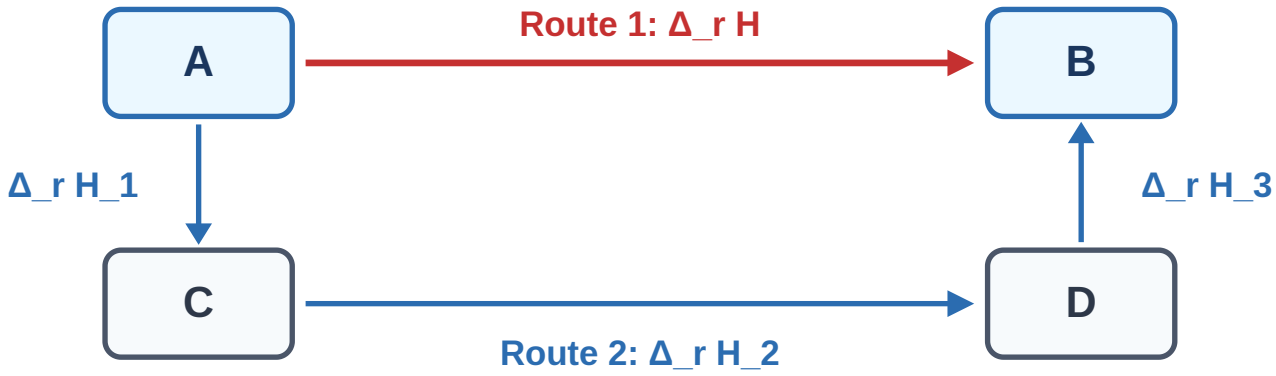
1. A balanced chemical equation together with its value of $\Delta_r H$ and physical states (including allotropic states) of all species is called a **thermochemical equation**.
2. The stoichiometric coefficients refer strictly to the **number of moles** (never molecules) of reactants and products.
3. The unit $\mathrm{kJ\,mol^{-1}}$ means *per mole of reaction as written*. If the stoichiometric coefficients of an equation are multiplied or divided by a factor, $\Delta_r H^\ominus$ is multiplied or divided by the same factor (showing enthalpy is extensive).
4. When a chemical equation is reversed, the magnitude of $\Delta_r H^\ominus$ remains the same, but its **sign is reversed**.

6.5 Hess's Law of Constant Heat Summation

Statement: *If a reaction takes place in several steps, then its standard reaction enthalpy is the sum of the standard enthalpies of the intermediate reactions into which the overall reaction may be divided at the same temperature.*

$$\Delta_r H = \Delta_r H_1 + \Delta_r H_2 + \Delta_r H_3 + \cdots$$

BOSONANCE



Tutorial Diagram 4: Hess's Law of Constant Heat Summation (Δ_r H = Δ_r H_1 + Δ_r H_2 + Δ_r H_3).

6.6 Enthalpies for Different Types of Reactions

- **Standard Enthalpy of Combustion ($\Delta_c H^\ominus$):** The enthalpy change per mole of a substance when it undergoes complete combustion with all reactants and products in their standard states at the specified temperature. Combustion reactions are exothermic ($\Delta_c H^\ominus < 0$).
  - Butane: $\mathrm{C_4H_{10}(g) + \frac{13}{2}O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l)}; \quad \Delta_c H^\ominus = -2658.0\,\mathrm{kJ\,mol^{-1}}$
  - Glucose: $\mathrm{C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)}; \quad \Delta_c H^\ominus = -2802.0\,\mathrm{kJ\,mol^{-1}}$
- **Enthalpy of Atomization ($\Delta_a H^\ominus$):** The enthalpy change on breaking $1\,\mathrm{mol}$ of a substance completely into isolated atoms in the gas phase.
  - For diatomic gases (e.g., $\mathrm{H_2(g) \rightarrow 2H(g)}$, $\Delta_a H^\ominus = 435.0\,\mathrm{kJ\,mol^{-1}}$), $\Delta_a H^\ominus$ equals the bond dissociation enthalpy.
  - For metals (e.g., $\mathrm{Na(s) \rightarrow Na(g)}$, $\Delta_a H^\ominus = 108.4\,\mathrm{kJ\,mol^{-1}}$), $\Delta_a H^\ominus$ equals the enthalpy of sublimation.
  - For methane: $\mathrm{CH_4(g) \rightarrow C(g) + 4H(g)}$, $\Delta_a H^\ominus = 1665\,\mathrm{kJ\,mol^{-1}}$.
- **Bond Enthalpy ($\Delta_{\mathrm{bond}} H^\ominus$):**
  - *Bond Dissociation Enthalpy:* Change in enthalpy when $1\,\mathrm{mol}$ of covalent bonds of a gaseous covalent compound is broken to form products in the gas phase (e.g., $\Delta_{\mathrm{Cl-Cl}} H^\ominus = 242\,\mathrm{kJ\,mol^{-1}}$, $\Delta_{\mathrm{O=O}} H^\ominus = 498\,\mathrm{kJ\,mol^{-1}}$).
  - *Mean Bond Enthalpy:* In polyatomic molecules like $\mathrm{CH_4}$, the four $\mathrm{C-H}$ bonds have identical length and energy, but successive bond dissociation steps require different energies ($+427$, $+439$, $+452$, and $+347\,\mathrm{kJ\,mol^{-1}}$). The average is the mean bond enthalpy: $\Delta_{\mathrm{C-H}} H^\ominus = \frac{1}{4}(1665) = 416\,\mathrm{kJ\,mol^{-1}}$.

$$\Delta_r H^\ominus \approx \sum(\text{Bond enthalpies of reactants}) - \sum(\text{Bond enthalpies of products})$$

Note: This relation is valid when all reactants and products are in the **gaseous state**.

Single Bond	Mean Bond Enthalpy ($\mathrm{kJ\,mol^{-1}}$)	Single Bond	Mean Bond Enthalpy ($\mathrm{kJ\,mol^{-1}}$)	Multiple Bond	Mean Bond Enthalpy ($\mathrm{kJ\,mol^{-1}}$)
$\mathrm{H-H}$	435.8	$\mathrm{C-C}$	347	$\mathrm{N=N}$	418
$\mathrm{C-H}$	414	$\mathrm{C-N}$	293	$\mathrm{N\equiv N}$	946
$\mathrm{N-H}$	389	$\mathrm{C-O}$	351	$\mathrm{C=N}$	615
$\mathrm{O-H}$	464	$\mathrm{C-Cl}$	330	$\mathrm{C\equiv N}$	891
$\mathrm{F-H}$	569	$\mathrm{N-N}$	159	$\mathrm{C=C}$	611
$\mathrm{Cl-H}$	431	$\mathrm{O-O}$	138	$\mathrm{C\equiv C}$	837
$\mathrm{Br-H}$	368	$\mathrm{F-F}$	155	$\mathrm{C=O}$	741
$\mathrm{I-H}$	297	$\mathrm{Cl-Cl}$	243	$\mathrm{C\equiv O}$	1070
$\mathrm{Br-Br}$	192	$\mathrm{I-I}$	151	$\mathrm{O=O}$	498

6.7 Lattice Enthalpy ($\Delta_{\mathrm{lattice}} H^\ominus$) and the Born-Haber Cycle

The **lattice enthalpy** of an ionic compound is the enthalpy change when $1\,\mathrm{mol}$ of an ionic compound dissociates completely into its ions in the gaseous state:

$$\mathrm{Na^+Cl^-(s) \rightarrow Na^+(g) + Cl^-(g)}; \quad \Delta_{\mathrm{lattice}} H^\ominus = +788\,\mathrm{kJ\,mol^{-1}}$$

Because lattice enthalpy cannot be measured directly by experiment, we construct a thermodynamic cycle based on Hess's law called the **Born-Haber Cycle**. For $\mathrm{NaCl(s)}$:

1. Sublimation of sodium: $\mathrm{Na(s) \rightarrow Na(g)}; \quad \Delta_{\mathrm{sub}} H^\ominus = +108.4\,\mathrm{kJ\,mol^{-1}}$
2. Ionization of sodium: $\mathrm{Na(g) \rightarrow Na^+(g) + e^-(g)}; \quad \Delta_i H^\ominus = +496\,\mathrm{kJ\,mol^{-1}}$

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3. Atomization of chlorine: $\frac{1}{2}\mathrm{Cl_2(g)} \rightarrow \mathrm{Cl(g)}$; $\quad \frac{1}{2}\Delta_{\mathrm{bond}}H^{\ominus} = +121\,\mathrm{kJ\,mol^{-1}}$
4. Electron gain by chlorine: $\mathrm{Cl(g) + e^-(g) \rightarrow Cl^-(g)}$; $\quad \Delta_{eg}H^{\ominus} = -348.6\,\mathrm{kJ\,mol^{-1}}$
5. Formation of $\mathrm{NaCl(s)}$: $\mathrm{Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)}$; $\quad \Delta_f H^{\ominus} = -411.2\,\mathrm{kJ\,mol^{-1}}$

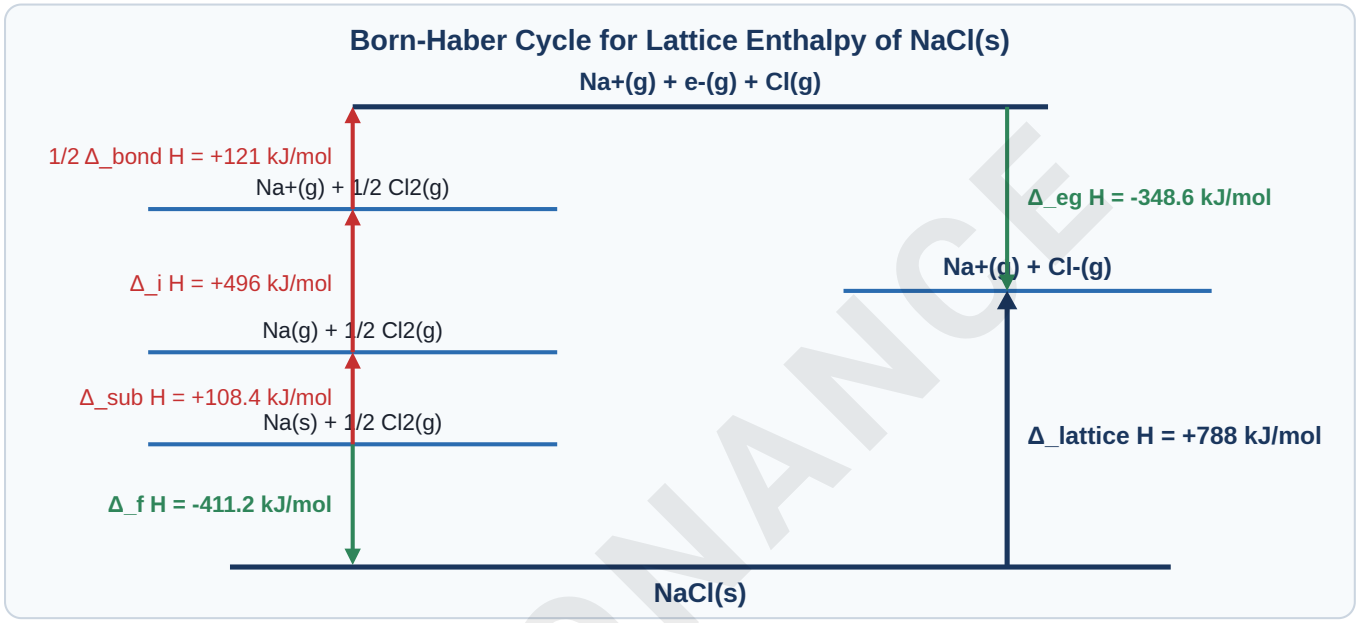
Applying Hess's law around the cycle for $\mathrm{NaCl(s) \rightarrow Na^+(g) + Cl^-(g)}$:

$$\Delta_{\mathrm{lattice}}H^{\ominus} = -\Delta_f H^{\ominus} + \Delta_{\mathrm{sub}}H^{\ominus} + \Delta_i H^{\ominus} + \frac{1}{2}\Delta_{\mathrm{bond}}H^{\ominus} + \Delta_{eg}H^{\ominus} = 411.2 + 108.4 + 496 + 121 - 348.6 = +788\,\mathrm{kJ\,mol^{-1}}$$

Since $\Delta n_g = 2$ for $\mathrm{NaCl(s) \rightarrow Na^+(g) + Cl^-(g)}$, the internal energy of lattice dissociation is smaller by $2RT$ ($\approx +783\,\mathrm{kJ\,mol^{-1}}$).

Relation at Temperature T vs. Absolute Zero: Ionization energy (E_0) and electron affinity (A) are defined at $0\,\mathrm{K}$. Taking $C_p = \frac{5}{2}R$ for each gaseous particle ($\mathrm{M(g), M^+(g), M^-(g), e^-(g)}$), $\Delta_r C_p^{\ominus} = +\frac{5}{2}R$ for ionization and $-\frac{5}{2}R$ for electron gain. Thus at temperature T:

$$\Delta_i H^{\ominus}(T) = E_0 + \frac{5}{2}RT \quad \text{and} \quad \Delta_{eg}H^{\ominus}(T) = -A - \frac{5}{2}RT$$



Tutorial Diagram 5: Born-Haber Enthalpy Diagram for Lattice Dissociation Enthalpy of NaCl(s).

6.8 Enthalpy of Solution ($\Delta_{\mathrm{sol}}H^{\ominus}$) and Enthalpy of Dilution

Enthalpy of Solution ($\Delta_{\mathrm{sol}}H^{\ominus}$): The enthalpy change when $1\,\mathrm{mol}$ of a solute dissolves in a specified amount of solvent (or in an infinite amount of solvent at infinite dilution, where ion-ion interactions are negligible). For an ionic solid $\mathrm{AB(s)}$ dissolving in water:

$$\Delta_{\mathrm{sol}}H^{\ominus} = \Delta_{\mathrm{lattice}}H^{\ominus} + \Delta_{\mathrm{hyd}}H^{\ominus}$$

For $\mathrm{NaCl(s)}$, $\Delta_{\mathrm{lattice}}H^{\ominus} = +788\,\mathrm{kJ\,mol^{-1}}$ and $\Delta_{\mathrm{hyd}}H^{\ominus} = -784\,\mathrm{kJ\,mol^{-1}}$, so $\Delta_{\mathrm{sol}}H^{\ominus} = +4\,\mathrm{kJ\,mol^{-1}}$. For most ionic compounds, $\Delta_{\mathrm{sol}}H^{\ominus} > 0$ (endothermic), so their solubility in water increases with rise in temperature.

Enthalpy of Dilution: The enthalpy change associated with adding additional solvent to a solution of specified original concentration at constant T and p. For example, given:

- $\mathrm{HCl(g) + 25\,aq. \rightarrow HCl \cdot 25\,aq.}$; $\quad \Delta H_1 = -72.03\,\mathrm{kJ\,mol^{-1}}$
- $\mathrm{HCl(g) + 40\,aq. \rightarrow HCl \cdot 40\,aq.}$; $\quad \Delta H_2 = -72.79\,\mathrm{kJ\,mol^{-1}}$

Subtracting the first from the second gives the enthalpy of dilution for $\mathrm{HCl \cdot 25\,aq. + 15\,aq. \rightarrow HCl \cdot 40\,aq.}$ as $\Delta H_{\mathrm{dil}} = -72.79 - (-72.03) = -0.76\,\mathrm{kJ\,mol^{-1}}$. At infinite dilution ($\mathrm{HCl \cdot \infty\,aq.}$), ΔH approaches $-74.85\,\mathrm{kJ\,mol^{-1}}$.

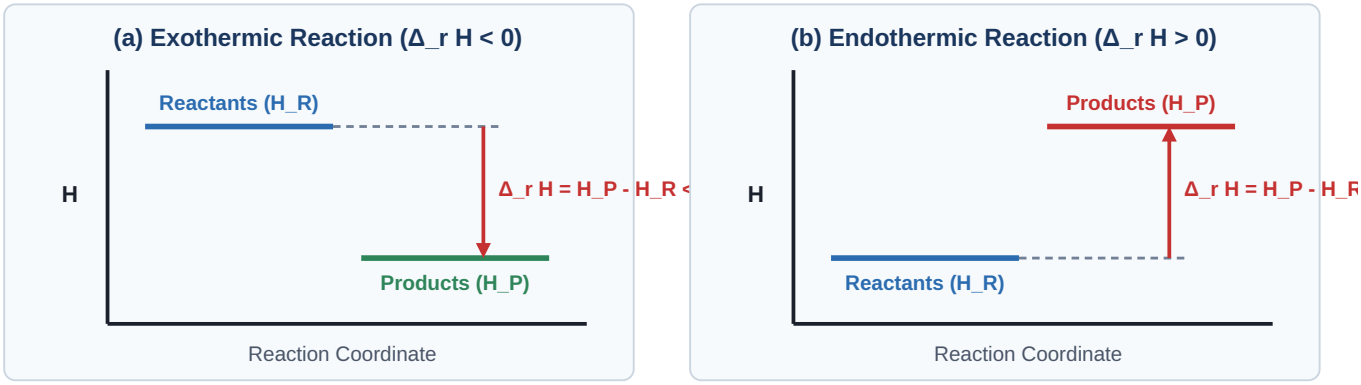
7. Spontaneity, Entropy (S), Gibbs Energy (G), and Equilibrium

7.1 What is a Spontaneous Process?

A **spontaneous process** is one that has the *potential to proceed without the assistance of an external agency*. It is unidirectional and irreversible on its own, and can only be reversed by an external agency. Spontaneity is a thermodynamic concept and **does not indicate the rate** of the reaction (for instance, $\mathrm{H_2}$ and $\mathrm{O_2}$ react spontaneously at room temperature, but at an imperceptibly slow rate).

While many spontaneous reactions are exothermic ($\Delta_r H^{\ominus} < 0$, such as formation of $\mathrm{NH_3}$, HCl, and $\mathrm{H_2O}$), several endothermic processes are also spontaneous—such as $\frac{1}{2}\mathrm{N_2(g) + O_2(g) \rightarrow NO_2(g)}$ ($\Delta_r H^{\ominus} = +33.2\,\mathrm{kJ\,mol^{-1}}$), $\mathrm{C(graphite, s) + 2S(l) \rightarrow CS_2(l)}$ ($\Delta_r H^{\ominus} = +128.5\,\mathrm{kJ\,mol^{-1}}$), melting of ice above $0\,^{\circ}\mathrm{C}$, or diffusion of two ideal gases into each other ($\Delta H = 0$). Thus, decrease in enthalpy is a contributory factor, not a universal criterion for spontaneity.

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Tutorial Diagram 6: Enthalpy Diagrams for (a) Exothermic (heat evolved) and (b) Endothermic (heat absorbed) Reactions.

7.2 Entropy (S) and the Second & Third Laws of Thermodynamics

Entropy (S) is a thermodynamic state function and extensive property that measures the **degree of randomness or disorder** in a system. For a given substance:

$$S_{\text{crystalline solid}} < S_{\text{liquid}} \ll S_{\text{gas}}$$

Heat added to a system increases molecular motion (randomness), and a given quantity of heat added at a *lower* temperature causes a greater relative increase in randomness than at a higher temperature. For a reversible process:

$$\Delta S_{\text{sys}} = \frac{q_{\text{rev}}}{T}$$

- **Second Law of Thermodynamics:** In any spontaneous process, the total entropy of the universe (system + surroundings) always increases:
  $$\Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0 \quad (\text{Spontaneous Process})$$
  At equilibrium, entropy reaches its maximum and $\Delta S_{\text{total}} = 0$.
- *Discriminating Reversible vs. Irreversible Isothermal Expansion of an Ideal Gas:* For both reversible and irreversible isothermal expansion of an ideal gas, $\Delta U = 0$. However, $\Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 0$ for a reversible process and $\Delta S_{\text{total}} > 0$ for an irreversible process (such as free expansion). Thus, ΔS_{total} **discriminates between reversible and irreversible processes, whereas ΔU does not**.
- **Third Law of Thermodynamics:** *The entropy of any pure crystalline substance approaches zero as the temperature approaches absolute zero ($0\,\text{K}$).* This allows the determination of **absolute entropies ($S^{\ominus}$)** of pure substances by summing $\frac{q_{\text{rev}}}{T}$ increments from $0\,\text{K}$ to $298\,\text{K}$. (Note: Solutions and supercooled liquids retain residual disorder and do not have zero entropy at $0\,\text{K}$).

7.3 Gibbs Energy (G) and Criterion of Spontaneity

To express spontaneity purely in terms of system properties at constant temperature and pressure, we define the **Gibbs energy (or Gibbs function, G)**:

$$G = H - TS \implies \Delta G_{\text{sys}} = \Delta H_{\text{sys}} - T\Delta S_{\text{sys}} \quad (\text{at constant } T \text{ and } p)$$

Since $\Delta S_{\text{surr}} = -\frac{\Delta H_{\text{sys}}}{T}$ at constant pressure, $\Delta S_{\text{total}} = \Delta S_{\text{sys}} - \frac{\Delta H_{\text{sys}}}{T}$, which gives:

$$T\Delta S_{\text{total}} = T\Delta S_{\text{sys}} - \Delta H_{\text{sys}} = -\Delta G_{\text{sys}}$$

Here, ΔH_{sys} is the total enthalpy change, $T\Delta S_{\text{sys}}$ is the energy unavailable to do useful work, and $-\Delta G$ **is the net free energy available to do useful (non-expansion) work**. At constant T and p:

- If $\Delta G < 0$ (negative): The process is **spontaneous**.
- If $\Delta G = 0$: The system is at **equilibrium**.
- If $\Delta G > 0$ (positive): The process is **non-spontaneous** in the forward direction (spontaneous in reverse).

Sign of $\Delta_r H^{\ominus}$	Sign of $\Delta_r S^{\ominus}$	Sign of $\Delta_r G^{\ominus} = \Delta_r H^{\ominus} - T\Delta_r S^{\ominus}$	Spontaneity Behaviour with Temperature
$-$ (Exothermic)	$+$	$-$ (Always negative)	**Spontaneous at all temperatures**
$-$ (Exothermic)	$-$	$-$ at low T (when $\lvert\Delta H\rvert > \lvert T\Delta S\rvert$) $+$ at high T (when $\lvert T\Delta S\rvert > \lvert\Delta H\rvert$)	**Spontaneous at low temperatures**; becomes non-spontaneous above $T = \Delta H/\Delta S$
$+$ (Endothermic)	$+$	$+$ at low T (when $\Delta H > T\Delta S$) $-$ at high T (when $T\Delta S > \Delta H$)	**Non-spontaneous at low temperatures; spontaneous at high temperatures** above $T = \Delta H/\Delta S$
$+$ (Endothermic)	$-$	$+$ (Always positive)	**Non-spontaneous at all temperatures**

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7.4 Gibbs Energy Change and Chemical Equilibrium

For a reversible chemical reaction $\mathrm{A} + \mathrm{B} \rightleftharpoons \mathrm{C} + \mathrm{D}$, dynamic equilibrium is reached when the Gibbs energy of the system is at a minimum ($\Delta_r G = 0$). The standard Gibbs energy change ($\Delta_r G^{\ominus}$) is related to the equilibrium constant (K) by:

$$\Delta_r G^{\ominus} = \Delta_r H^{\ominus} - T\Delta_r S^{\ominus} = -RT\ln K = -2.303\,RT\log_{10} K$$

- If $\Delta_r G^{\ominus} < 0$, then $\ln K > 0 \implies K > 1$ (products predominate at equilibrium).
- If $\Delta_r G^{\ominus} = 0$, then $\ln K = 0 \implies K = 1$.
- If $\Delta_r G^{\ominus} > 0$, then $\ln K < 0 \implies K < 1$ (reactants predominate at equilibrium).

8. Comprehensive Worked Examples

Study these step-by-step solved problems carefully to master numerical and conceptual applications for NEET UG.

Worked Example 1: First Law & Nature of System Boundaries

Conceptual Application

Problem: Express the change in internal energy (ΔU) of a system and identify the type of wall or system when:
(i) No heat is absorbed by the system from the surroundings, but work (w) is done on the system.
(ii) No work is done on the system, but q amount of heat is taken out from the system and given to the surroundings.
(iii) w amount of work is done by the system and q amount of heat is supplied to the system.

Solution:

- **(i)** Here $q = 0$ and work is done *on* the system ($+w$). From $\Delta U = q + w$, we get $\Delta U = w_{\text{ad}}$. Since no heat is permitted to cross the boundary, the wall is an **adiabatic wall**.
- **(ii)** Here $w = 0$ and heat is lost by the system ($-q$). Thus $\Delta U = -q$. Since heat transfers across the boundary, the system has **thermally conducting walls**.
- **(iii)** Heat supplied to the system is $+q$ and work done *by* the system is $-w$. Thus $\Delta U = q - w$. Since energy is exchanged as both heat and work without transfer of matter, it is a **closed system**.

Worked Example 2: Isothermal Expansion — Vacuum vs. Irreversible vs. Reversible

Ideal Gas Work

Problem: $1\,\text{mol}$ of an ideal gas initially occupying $2\,\text{L}$ at a pressure of $10\,\text{atm}$ expands isothermally at $25\,^{\circ}\text{C}$ ($298\,\text{K}$) until its final volume is $10\,\text{L}$. Calculate the heat absorbed (q) and work done (w) when the expansion occurs:
(a) Into a vacuum (free expansion),
(b) Against a constant external pressure of $1\,\text{atm}$, and
(c) Reversibly.

Solution: Since the process is isothermal ($T = 298\,\text{K} = \text{constant}$) for an ideal gas, $\Delta U = 0$, so $q = -w$.

- **(a) Expansion into vacuum ($p_{\text{ex}} = 0$):**
$$w = -p_{\text{ex}}(V_f - V_i) = -0 \times (10 - 2) = 0\,\text{L atm}, \quad q = 0\,\text{L atm}$$
No work is done and no heat is absorbed.
- **(b) Against constant external pressure ($p_{\text{ex}} = 1\,\text{atm}$):**
$$w = -p_{\text{ex}}(V_f - V_i) = -1\,\text{atm} \times (10 - 2)\,\text{L} = -8\,\text{L atm}$$
$$q = -w = +8\,\text{L atm}$$
- **(c) Reversible isothermal expansion:**
$$w_{\text{rev}} = -2.303\,nRT\log_{10}\left(\frac{V_f}{V_i}\right) = -2.303 \times (1\,\text{mol}) \times (0.08206\,\text{L atm K}^{-1}\,\text{mol}^{-1}) \times (298\,\text{K}) \times \log_{10}\left(\frac{10}{2}\right)$$
$$w_{\text{rev}} = -2.303 \times 0.08206 \times 298 \times 0.6990 = -39.366\,\text{L atm}$$
$$q = -w_{\text{rev}} = +39.366\,\text{L atm}$$
Key Takeaway: Work done by the gas during reversible expansion ($39.366\,\text{L atm}$) is significantly greater than during single-step irreversible expansion ($8\,\text{L atm}$).

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Worked Example 3: Relating ΔH and ΔU for Phase Change and Combustion

Enthalpy & Internal Energy

Problem:
(a) A swimmer coming out from a pool is covered with a film of water weighing $18\,\text{g}$. How much heat must be supplied to evaporate this water at $298\,\text{K}$? Also calculate the internal energy of vaporization ($\Delta_{\text{vap}}U^{\ominus}$) at $298\,\text{K}$, given $\Delta_{\text{vap}}H^{\ominus} = 44.01\,\text{kJ mol}^{-1}$ at $298\,\text{K}$.
(b) The reaction of solid cyanamide, $\text{NH}_2\text{CN(s)}$, with dioxygen was carried out in a bomb calorimeter, and ΔU was found to be $-742.7\,\text{kJ mol}^{-1}$ at $298\,\text{K}$. Calculate $\Delta H^{\ominus}$ at $298\,\text{K}$ for the reaction:

$$\text{NH}_2\text{CN(s)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + \text{H}_2\text{O(l)}$$

Solution (a):
Moles of water in $18\,\text{g}$: $n = \dfrac{18\,\text{g}}{18\,\text{g mol}^{-1}} = 1\,\text{mol}.$
For the evaporation $\text{H}_2\text{O(l)} \rightarrow \text{H}_2\text{O(g)}$, heat supplied at constant pressure is:

$$q_p = n\Delta_{\text{vap}}H^{\ominus} = 1\,\text{mol} \times 44.01\,\text{kJ mol}^{-1} = 44.01\,\text{kJ}$$

Here $\Delta n_g = 1 - 0 = 1$. Using $\Delta_{\text{vap}}U^{\ominus} = \Delta_{\text{vap}}H^{\ominus} - \Delta n_g RT$:

$$\Delta_{\text{vap}}U^{\ominus} = 44.01\,\text{kJ mol}^{-1} - (1) \times (8.314 \times 10^{-3}\,\text{kJ K}^{-1}\,\text{mol}^{-1}) \times (298\,\text{K}) = 44.01 - 2.48 = 41.53\,\text{kJ mol}^{-1}$$

Solution (b):
In $\text{NH}_2\text{CN(s)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + \text{H}_2\text{O(l)}$:
Moles of gaseous products $= 1 + 1 = 2\,\text{mol}$; moles of gaseous reactants $= \frac{3}{2} = 1.5\,\text{mol}$.
Therefore, $\Delta n_g = 2 - 1.5 = +0.5\,\text{mol}$.

$$\Delta H^{\ominus} = \Delta U^{\ominus} + \Delta n_g RT = -742.7\,\text{kJ mol}^{-1} + (0.5) \times (8.314 \times 10^{-3}\,\text{kJ K}^{-1}\,\text{mol}^{-1}) \times (298\,\text{K})$$

$$\Delta H^{\ominus} = -742.7 + 1.24 = -741.5\,\text{kJ mol}^{-1}$$

Worked Example 4: Bomb Calorimetry Calculation

Calorimetry

Problem: $1.0\,\text{g}$ of graphite is burnt in a bomb calorimeter in excess of oxygen at $298\,\text{K}$ and $1\,\text{atm}$ pressure according to $\text{C(graphite)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$. During the reaction, the temperature rises from $298\,\text{K}$ to $299\,\text{K}$. If the heat capacity of the bomb calorimeter is $20.7\,\text{kJ K}^{-1}$, what is the molar enthalpy change (ΔH) for the reaction at $298\,\text{K}$ and $1\,\text{atm}$?

Solution:
Heat evolved by combustion of $1.0\,\text{g}$ of graphite is absorbed by the calorimeter:

$$q_V = -C_V \times \Delta T = -20.7\,\text{kJ K}^{-1} \times (299 - 298)\,\text{K} = -20.7\,\text{kJ}$$

For $1\,\text{mol}$ $(12.0\,\text{g})$ of graphite, the internal energy change is:

$$\Delta U = \frac{-20.7\,\text{kJ}}{1.0\,\text{g}} \times 12.0\,\text{g mol}^{-1} = -248.4\,\text{kJ mol}^{-1} \approx -2.48 \times 10^2\,\text{kJ mol}^{-1}$$

For $\text{C(graphite, s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$, $\Delta n_g = 1 - 1 = 0$. Therefore:

$$\Delta H = \Delta U + \Delta n_g RT = \Delta U = -2.48 \times 10^2\,\text{kJ mol}^{-1}$$

Worked Example 5: Multi-Step Cooling and Phase Transformation

Phase Transformation & Heat Capacity

Problem: Calculate the total enthalpy change on freezing $1.0\,\text{mol}$ of water at $10.0\,^{\circ}\text{C}$ to ice at $-10.0\,^{\circ}\text{C}$. Given: $\Delta_{\text{fus}}H = 6.03\,\text{kJ mol}^{-1}$ at $0\,^{\circ}\text{C}$, $C_p[\text{H}_2\text{O(l)}] = 75.3\,\text{J mol}^{-1}\,\text{K}^{-1}$, and $C_p[\text{H}_2\text{O(s)}] = 36.8\,\text{J mol}^{-1}\,\text{K}^{-1}$.

Solution: By Hess's law, divide the overall change into three sequential steps:

1. **Step 1 — Cooling $1.0\,\text{mol}$ liquid water from $10.0\,^{\circ}\text{C}$ to $0\,^{\circ}\text{C}$ ($\Delta T = -10.0\,\text{K}$):**
$$\Delta H_1 = nC_p[\text{H}_2\text{O(l)}]\Delta T = 1.0\,\text{mol} \times 75.3\,\text{J mol}^{-1}\,\text{K}^{-1} \times (-10.0\,\text{K}) = -753\,\text{J mol}^{-1} = -0.753\,\text{kJ mol}^{-1}$$
2. **Step 2 — Freezing $1.0\,\text{mol}$ liquid water to ice at $0\,^{\circ}\text{C}$:**
$$\Delta H_2 = -\Delta_{\text{fus}}H = -6.03\,\text{kJ mol}^{-1}$$
3. **Step 3 — Cooling $1.0\,\text{mol}$ solid ice from $0\,^{\circ}\text{C}$ to $-10.0\,^{\circ}\text{C}$ ($\Delta T = -10.0\,\text{K}$):**
$$\Delta H_3 = nC_p[\text{H}_2\text{O(s)}]\Delta T = 1.0\,\text{mol} \times 36.8\,\text{J mol}^{-1}\,\text{K}^{-1} \times (-10.0\,\text{K}) = -368\,\text{J mol}^{-1} = -0.368\,\text{kJ mol}^{-1}$$

Total Enthalpy Change:

$$\Delta H_{\text{total}} = \Delta H_1 + \Delta H_2 + \Delta H_3 = -0.753 - 6.03 - 0.368 = -7.151\,\text{kJ mol}^{-1}$$

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Worked Example 6: Hess's Law, Enthalpy of Formation, and Mean Bond Enthalpy

Thermochemistry

Problem:

(a) The combustion of $1\ \text{mol}$ of liquid benzene takes place at $298\ \text{K}$ and $1\ \text{atm}$ to give $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$, liberating $3267.0\ \text{kJ}$ of heat. Given $\Delta_f H^\ominus[\text{CO}_2(\text{g})] = -393.5\ \text{kJ mol}^{-1}$ and $\Delta_f H^\ominus[\text{H}_2\text{O}(\text{l})] = -285.83\ \text{kJ mol}^{-1}$, calculate $\Delta_f H^\ominus$ of liquid benzene.

(b) Calculate the enthalpy change for the process $\text{CCl}_4(\text{g}) \rightarrow \text{C}(\text{g}) + 4\text{Cl}(\text{g})$ and find the mean $\text{C}-\text{Cl}$ bond enthalpy in $\text{CCl}_4(\text{g})$. Given: $\Delta_{\text{vap}} H^\ominus[\text{CCl}_4(\text{l})] = 30.5\ \text{kJ mol}^{-1}$, $\Delta_f H^\ominus[\text{CCl}_4(\text{l})] = -135.5\ \text{kJ mol}^{-1}$, $\Delta_a H^\ominus[\text{C(graphite)}] = 715.0\ \text{kJ mol}^{-1}$, and $\Delta_a H^\ominus[\text{Cl}_2(\text{g})] = 242\ \text{kJ mol}^{-1}$.

Solution (a):

Combustion equation: $\text{C}_6\text{H}_6(\text{l}) + \frac{15}{2}\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l})$; $\quad \Delta_c H^\ominus = -3267.0\ \text{kJ mol}^{-1}$.

Using $\Delta_c H^\ominus = [6\Delta_f H^\ominus(\text{CO}_2) + 3\Delta_f H^\ominus(\text{H}_2\text{O})] - \Delta_f H^\ominus(\text{C}_6\text{H}_6)$:

$$-3267.0 = [6(-393.5) + 3(-285.83)] - \Delta_f H^\ominus(\text{C}_6\text{H}_6) = (-2361.0 - 857.49) - \Delta_f H^\ominus(\text{C}_6\text{H}_6)$$

$$\Delta_f H^\ominus[\text{C}_6\text{H}_6(\text{l})] = -3218.49 + 3267.0 = +48.51\ \text{kJ mol}^{-1}$$

Solution (b):

Write the thermochemical equations:

(1) $\text{C(graphite)} + 2\text{Cl}_2(\text{g}) \rightarrow \text{CCl}_4(\text{l})$; $\quad \Delta_f H^\ominus = -135.5\ \text{kJ mol}^{-1}$

(2) $\text{CCl}_4(\text{l}) \rightarrow \text{CCl}_4(\text{g})$; $\quad \Delta_{\text{vap}} H^\ominus = +30.5\ \text{kJ mol}^{-1}$

Adding (1) and (2) gives $\Delta_f H^\ominus[\text{CCl}_4(\text{g})] = -135.5 + 30.5 = -105.0\ \text{kJ mol}^{-1}$.

Now, for $\text{CCl}_4(\text{g}) \rightarrow \text{C}(\text{g}) + 4\text{Cl}(\text{g})$:

$$\Delta_a H^\ominus = \Delta_a H^\ominus(\text{C}) + 2\Delta_a H^\ominus(\text{Cl}_2) - \Delta_f H^\ominus[\text{CCl}_4(\text{g})] = 715.0 + 2(242) - (-105.0) = 1304.0\ \text{kJ mol}^{-1}$$

Since $4\ \text{mol}$ of $\text{C}-\text{Cl}$ bonds are broken per mole of $\text{CCl}_4(\text{g})$:

$$\text{Mean C}-\text{Cl Bond Enthalpy} = \frac{1304.0\ \text{kJ mol}^{-1}}{4} = 326.0\ \text{kJ mol}^{-1}$$

Worked Example 7: Entropy of System & Surroundings and Spontaneity

Second Law & Gibbs Energy

Problem:

(a) For the oxidation of iron, $4\text{Fe}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{Fe}_2\text{O}_3(\text{s})$, the entropy change of the system ($\Delta_r S^\ominus_{\text{sys}}$) is $-549.4\ \text{J K}^{-1}\ \text{mol}^{-1}$ at $298\ \text{K}$, and $\Delta_r H^\ominus = -1648 \times 10^3\ \text{J mol}^{-1}$. In spite of the negative entropy change of the system, why is the reaction spontaneous?

(b) For the reaction $2\text{A}(\text{g}) + \text{B}(\text{g}) \rightarrow 2\text{D}(\text{g})$ at $298\ \text{K}$, $\Delta U^\ominus = -10.5\ \text{kJ}$ and $\Delta S^\ominus = -44.1\ \text{J K}^{-1}$. Calculate $\Delta G^\ominus$ for the reaction and predict whether it can occur spontaneously.

Solution (a):

Spontaneity is determined by $\Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$. At constant pressure and $T = 298\ \text{K}$:

$$\Delta S_{\text{surr}} = -\frac{\Delta_r H^\ominus}{T} = -\frac{-1648 \times 10^3\ \text{J mol}^{-1}}{298\ \text{K}} = +5530.2\ \text{J K}^{-1}\ \text{mol}^{-1}$$

$$\Delta S_{\text{total}} = -549.4 + 5530.2 = +4980.8\ \text{J K}^{-1}\ \text{mol}^{-1} > 0$$

Because ΔS_{total} is positive, the reaction is spontaneous.

Solution (b):

For $2\text{A}(\text{g}) + \text{B}(\text{g}) \rightarrow 2\text{D}(\text{g})$, $\Delta n_g = 2 - (2 + 1) = -1$.

First calculate $\Delta H^\ominus$ at $T = 298\ \text{K}$:

$$\Delta H^\ominus = \Delta U^\ominus + \Delta n_g RT = -10.5\ \text{kJ} + (-1)(8.314 \times 10^{-3}\ \text{kJ K}^{-1}\ \text{mol}^{-1})(298\ \text{K}) = -10.5 - 2.48 = -12.98\ \text{kJ}$$

Now calculate $\Delta G^\ominus$:

$$\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus = -12.98\ \text{kJ} - (298\ \text{K})(-44.1 \times 10^{-3}\ \text{kJ K}^{-1}) = -12.98 + 13.14 = +0.16\ \text{kJ}$$

Since $\Delta G^\ominus = +0.16\ \text{kJ} > 0$ (positive), the reaction is **non-spontaneous** at $298\ \text{K}$.

BOSONANCE

Worked Example 8: Equilibrium Constant (K_p) and Standard Gibbs Energy ($\Delta_r G^\ominus$)

Chemical Equilibrium

Problem:
(a) Find the equilibrium constant K at $298\,\text{K}$ for the reaction $2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g}) \rightleftharpoons \text{NH}_2\text{CONH}_2(\text{aq}) + \text{H}_2\text{O}(\text{l})$, given $\Delta_r G^\ominus = -13.6\,\text{kJ mol}^{-1}$.
(b) At $60\,^\circ\text{C}$ ($333\,\text{K}$), dinitrogen tetroxide (N_2O_4) is 50% dissociated according to $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ at a total pressure of $1\,\text{atm}$. Calculate the standard free energy change ($\Delta_r G^\ominus$) at this temperature.

Solution (a):

$$\log_{10} K = \frac{-\Delta_r G^\ominus}{2.303\,RT} = \frac{-(-13.6 \times 10^3\,\text{J mol}^{-1})}{2.303 \times 8.314\,\text{J K}^{-1}\,\text{mol}^{-1} \times 298\,\text{K}} = 2.38$$

$$K = \text{antilog}(2.38) = 2.4 \times 10^2$$

Solution (b):
Start with $1\,\text{mol}$ of $\text{N}_2\text{O}_4(\text{g})$. At 50% dissociation ($\alpha = 0.5$):
Moles at equilibrium: $n(\text{N}_2\text{O}_4) = 1 - 0.5 = 0.5\,\text{mol}$; $n(\text{NO}_2) = 2 \times 0.5 = 1.0\,\text{mol}$; total moles $= 1.5\,\text{mol}$.
Partial pressures at $p_{\text{total}} = 1\,\text{atm}$:

$$p_{\text{N}_2\text{O}_4} = \frac{0.5}{1.5} \times 1\,\text{atm} = \frac{1}{3}\,\text{atm}, \quad p_{\text{NO}_2} = \frac{1.0}{1.5} \times 1\,\text{atm} = \frac{2}{3}\,\text{atm}$$

$$K_p = \frac{(p_{\text{NO}_2})^2}{p_{\text{N}_2\text{O}_4}} = \frac{(2/3)^2}{1/3} = \frac{4}{3} = 1.33\,\text{atm}$$

$$\Delta_r G^\ominus = -2.303\,RT \log_{10} K_p = -2.303 \times (8.314\,\text{J K}^{-1}\,\text{mol}^{-1}) \times (333\,\text{K}) \times \log_{10}(1.33)$$

$$\Delta_r G^\ominus = -6375.9 \times 0.1239 = -789.9\,\text{J mol}^{-1} \approx -0.79\,\text{kJ mol}^{-1}$$

9. Original NEET UG Practice Questions with Worked Solutions

Note: All questions below are original practice questions designed strictly in accordance with the NEET UG syllabus and examination pattern (including single-correct MCQs, multi-statement evaluations, Assertion-Reason, and Matrix Match formats).

Practice Question 1

NEET UG — Single Choice MCQ

Which of the following sets contains **only** thermodynamic state functions and intensive properties?

(1) Internal energy (U), Molar enthalpy (H_m), Temperature (T)

(2) Pressure (p), Temperature (T), Molar heat capacity (C_m)

(3) Specific heat capacity (c), Work (w), Density (ρ)

(4) Enthalpy (H), Entropy (S), Gibbs energy (G)

Correct Answer: (2)

Worked Solution:
State functions depend only on the state of the system, and intensive properties are independent of the amount of matter present.
• In (1), U is a state function but an *extensive* property.
• In (2), pressure (p), temperature (T), and molar heat capacity ($C_\text{m} = C/n$) are all state functions (or state variables) and are all **intensive** properties.
• In (3), work (w) is a path function and extensive.
• In (4), H, S, and G are state functions, but all three are extensive properties.

BOSONANCE

Practice Question 2 NEET UG — Single Choice MCQ

In a closed thermodynamic process, a system absorbs $701\ \text{J}$ of heat from the surroundings while $394\ \text{J}$ of mechanical work is done by the system on the surroundings. What is the change in internal energy (ΔU) for the process?

(1) $-1095\ \text{J}$

(2) $+1095\ \text{J}$

(3) $+307\ \text{J}$

(4) $-307\ \text{J}$

Correct Answer: (3)

Worked Solution:
According to IUPAC sign conventions:
• Heat absorbed by the system: $q = +701\ \text{J}$
• Work done *by* the system: $w = -394\ \text{J}$
By the First Law of Thermodynamics:

$$\Delta U = q + w = +701\ \text{J} + (-394\ \text{J}) = +307\ \text{J}$$

Practice Question 3 NEET UG — Single Choice MCQ

The standard internal energy change ($\Delta U^{\ominus}$) for the complete combustion of methane, $\text{CH}_4(\text{g})$, to $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$ at $298\ \text{K}$ is $-X\ \text{kJ mol}^{-1}$. What is the correct relation for the standard enthalpy change ($\Delta H^{\ominus}$) of this reaction?

(1) $\Delta H^{\ominus} = \Delta U^{\ominus}$

(2) $\Delta H^{\ominus} > \Delta U^{\ominus}$

(3) $\Delta H^{\ominus} < \Delta U^{\ominus}$

(4) $\Delta H^{\ominus} = 0$

Correct Answer: (3)

Worked Solution:
The balanced equation for the combustion of methane at $298\ \text{K}$ is:

$$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$$

Here, $\Delta n_g = n_g(\text{products}) - n_g(\text{reactants}) = 1 - (1 + 2) = -2\ \text{mol}$.
Since $\Delta H^{\ominus} = \Delta U^{\ominus} + \Delta n_g RT = -X - 2RT$, $\Delta H^{\ominus}$ is more negative than $\Delta U^{\ominus}$. Therefore, $\Delta H^{\ominus} < \Delta U^{\ominus}$.

BOSONANCE

Practice Question 4 NEET UG — Numerical MCQ

How many kilojoules of heat are required to raise the temperature of $60.0\,\text{g}$ of aluminium from $35\,^\circ\text{C}$ to $55\,^\circ\text{C}$? (Given: Molar mass of $\text{Al} = 27.0\,\text{g mol}^{-1}$; Molar heat capacity of $\text{Al} = 24\,\text{J mol}^{-1}\,\text{K}^{-1}$).

(1) $1.07\,\text{kJ}$

(2) $2.40\,\text{kJ}$

(3) $0.48\,\text{kJ}$

(4) $28.8\,\text{kJ}$

Correct Answer: (1)

Worked Solution:

Number of moles of aluminium: $n = \dfrac{60.0\,\text{g}}{27.0\,\text{g mol}^{-1}} = \dfrac{20}{9}\,\text{mol} \approx 2.222\,\text{mol}.$

Temperature change: $\Delta T = 55\,^\circ\text{C} - 35\,^\circ\text{C} = 20\,\text{K}.$

Heat required:

$$q = nC_\text{m}\Delta T = \left(\frac{20}{9}\,\text{mol}\right) \times \left(24\,\text{J mol}^{-1}\,\text{K}^{-1}\right) \times (20\,\text{K}) = 1066.7\,\text{J} \approx 1.07\,\text{kJ}$$

Practice Question 5 NEET UG — Numerical MCQ

The standard enthalpies of combustion of methane ($\text{CH}_4(\text{g})$), graphite ($\text{C}(\text{graphite, s})$), and dihydrogen ($\text{H}_2(\text{g})$) at $298\,\text{K}$ are $-890.3\,\text{kJ mol}^{-1}$, $-393.5\,\text{kJ mol}^{-1}$, and $-285.8\,\text{kJ mol}^{-1}$, respectively. The standard enthalpy of formation ($\Delta_f H^\ominus$) of $\text{CH}_4(\text{g})$ at $298\,\text{K}$ is:

(1) $-74.8\,\text{kJ mol}^{-1}$

(2) $-52.27\,\text{kJ mol}^{-1}$

(3) $+74.8\,\text{kJ mol}^{-1}$

(4) $+52.26\,\text{kJ mol}^{-1}$

Correct Answer: (1)

Worked Solution:

Required formation equation: $\text{C}(\text{graphite, s}) + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g}).$

Given combustion data:

(i) $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}); \quad \Delta_c H_1^\ominus = -890.3\,\text{kJ mol}^{-1}$

(ii) $\text{C}(\text{graphite, s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}); \quad \Delta_c H_2^\ominus = -393.5\,\text{kJ mol}^{-1}$

(iii) $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l}); \quad \Delta_c H_3^\ominus = -285.8\,\text{kJ mol}^{-1}$

Applying Hess's law: $\text{Eq. (ii)} + 2 \times \text{Eq. (iii)} - \text{Eq. (i)}$:

$$\Delta_f H^\ominus[\text{CH}_4(\text{g})] = -393.5 + 2(-285.8) - (-890.3) = -393.5 - 571.6 + 890.3 = -74.8\,\text{kJ mol}^{-1}$$

BOSONANCE

Practice Question 6 NEET UG — Numerical MCQ

The standard enthalpies of formation of $\mathrm{CO(g)}$, $\mathrm{CO_2(g)}$, $\mathrm{N_2O(g)}$, and $\mathrm{N_2O_4(g)}$ are -110, -393, $+81$, and $+9.7\,\mathrm{kJ\,mol^{-1}}$, respectively. What is the value of $\Delta_r H^\ominus$ for the reaction:

$$\mathrm{N_2O_4(g) + 3CO(g) \rightarrow N_2O(g) + 3CO_2(g)}$$

(1) $-777.7\,\mathrm{kJ\,mol^{-1}}$

(2) $+777.7\,\mathrm{kJ\,mol^{-1}}$

(3) $-858.7\,\mathrm{kJ\,mol^{-1}}$

(4) $-211.7\,\mathrm{kJ\,mol^{-1}}$

Correct Answer: (1)

Worked Solution:

Using $\Delta_r H^\ominus = \sum a_i \Delta_f H^\ominus(\text{products}) - \sum b_i \Delta_f H^\ominus(\text{reactants})$:

$$\Delta_r H^\ominus = \left[\Delta_f H^\ominus(\mathrm{N_2O}) + 3\Delta_f H^\ominus(\mathrm{CO_2})\right] - \left[\Delta_f H^\ominus(\mathrm{N_2O_4}) + 3\Delta_f H^\ominus(\mathrm{CO})\right]$$

$$\Delta_r H^\ominus = [81 + 3(-393)] - [9.7 + 3(-110)] = (81 - 1179) - (9.7 - 330) = -1098 - (-320.3) = -777.7\,\mathrm{kJ\,mol^{-1}}$$

Practice Question 7 NEET UG — Numerical MCQ

The enthalpy of combustion of carbon to $\mathrm{CO_2(g)}$ is $-393.5\,\mathrm{kJ\,mol^{-1}}$. What is the magnitude of heat released upon the formation of $35.2\,\mathrm{g}$ of $\mathrm{CO_2}$ from carbon and dioxygen gas?

(1) $393.5\,\mathrm{kJ}$

(2) $314.8\,\mathrm{kJ}$

(3) $196.75\,\mathrm{kJ}$

(4) $491.9\,\mathrm{kJ}$

Correct Answer: (2)

Worked Solution:

Molar mass of $\mathrm{CO_2} = 12.0 + 2(16.0) = 44.0\,\mathrm{g\,mol^{-1}}$.

Moles of $\mathrm{CO_2}$ formed: $n = \dfrac{35.2\,\mathrm{g}}{44.0\,\mathrm{g\,mol^{-1}}} = 0.80\,\mathrm{mol}$.

Since formation of $1.0\,\mathrm{mol}$ of $\mathrm{CO_2}$ releases $393.5\,\mathrm{kJ}$ of heat, formation of $0.80\,\mathrm{mol}$ releases:

$$q = 0.80\,\mathrm{mol} \times 393.5\,\mathrm{kJ\,mol^{-1}} = 314.8\,\mathrm{kJ}$$

BOSONANCE

Practice Question 8 NEET UG — Numerical MCQ

Calculate the standard enthalpy of formation ($\Delta_f H^\ominus$) of liquid methanol, $\mathrm{CH_3OH(l)}$, from the following data at $298\ \mathrm{K}$:
(i) $\mathrm{CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)}$; $\quad \Delta_r H^\ominus = -726\ \mathrm{kJ\ mol^{-1}}$
(ii) $\mathrm{C(graphite, s) + O_2(g) \rightarrow CO_2(g)}$; $\quad \Delta_c H^\ominus = -393\ \mathrm{kJ\ mol^{-1}}$
(iii) $\mathrm{H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)}$; $\quad \Delta_f H^\ominus = -286\ \mathrm{kJ\ mol^{-1}}$

(1) $-239\ \mathrm{kJ\ mol^{-1}}$

(2) $+239\ \mathrm{kJ\ mol^{-1}}$

(3) $-1691\ \mathrm{kJ\ mol^{-1}}$

(4) $-47\ \mathrm{kJ\ mol^{-1}}$

Correct Answer: (1)

Worked Solution:
Target formation equation: $\mathrm{C(graphite, s) + 2H_2(g) + \frac{1}{2}O_2(g) \rightarrow CH_3OH(l)}$.
Combine the given equations as $\mathrm{Eq.\ (ii) + 2 \times Eq.\ (iii) - Eq.\ (i)}$:

$$\Delta_f H^\ominus[\mathrm{CH_3OH(l)}] = -393 + 2(-286) - (-726) = -393 - 572 + 726 = -239\ \mathrm{kJ\ mol^{-1}}$$

Practice Question 9 NEET UG — Conceptual MCQ

For the chemical reaction $\mathrm{2Cl(g) \rightarrow Cl_2(g)}$, what are the correct signs of ΔH and ΔS?

(1) $\Delta H > 0$ and $\Delta S > 0$

(2) $\Delta H < 0$ and $\Delta S < 0$

(3) $\Delta H < 0$ and $\Delta S > 0$

(4) $\Delta H > 0$ and $\Delta S < 0$

Correct Answer: (2)

Worked Solution:
• **Sign of ΔH:** The reaction $\mathrm{2Cl(g) \rightarrow Cl_2(g)}$ represents the formation of a covalent $\mathrm{Cl-Cl}$ bond from isolated gaseous chlorine atoms. Bond formation always releases energy (exothermic), so $\Delta H < 0$ (negative).
• **Sign of ΔS:** Two moles of gaseous atoms combine to form one mole of gaseous diatomic molecules ($\Delta n_g = 1 - 2 = -1$). Decrease in the number of gaseous particles reduces randomness, so $\Delta S < 0$ (negative).

BOSONANCE

Practice Question 10 NEET UG — Numerical MCQ

For the reaction $2\mathrm{A} + \mathrm{B} \rightarrow \mathrm{C}$ at $298\,\mathrm{K}$, $\Delta H = +400\,\mathrm{kJ\,mol^{-1}}$ and $\Delta S = +0.2\,\mathrm{kJ\,K^{-1}\,mol^{-1}}$. Assuming ΔH and ΔS remain constant over the temperature range, at what temperature will the reaction become spontaneous?

(1) Below $2000\,\mathrm{K}$

(2) At $298\,\mathrm{K}$

(3) Above $2000\,\mathrm{K}$

(4) Below $500\,\mathrm{K}$

Correct Answer: (3)

Worked Solution:
First find the equilibrium crossover temperature where $\Delta G = 0$:

$$\Delta G = \Delta H - T\Delta S = 0 \implies T_{\text{eq}} = \frac{\Delta H}{\Delta S} = \frac{400\,\mathrm{kJ\,mol^{-1}}}{0.2\,\mathrm{kJ\,K^{-1}\,mol^{-1}}} = 2000\,\mathrm{K}$$

Since both $\Delta H > 0$ and $\Delta S > 0$, $\Delta G = \Delta H - T\Delta S$ becomes negative (< 0) only when $T\Delta S > \Delta H$, which requires $T > 2000\,\mathrm{K}$.

Practice Question 11 NEET UG — Conceptual MCQ

A chemical reaction $\mathrm{A} + \mathrm{B} \rightarrow \mathrm{C} + \mathrm{D} + q$ (where $q > 0$ represents heat evolved) is found to have a positive entropy change ($\Delta S > 0$). The reaction will be:

(1) Spontaneous only at high temperatures

(2) Spontaneous only at low temperatures

(3) Non-spontaneous at any temperature

(4) Spontaneous at all temperatures

Correct Answer: (4)

Worked Solution:
Because heat q is evolved as a product, the reaction is exothermic ($\Delta H < 0$). We are also given that $\Delta S > 0$.
From $\Delta G = \Delta H - T\Delta S$, since ΔH is negative and $-T\Delta S$ is also negative for all $T > 0\,\mathrm{K}$, ΔG is negative at all temperatures. Hence the reaction is spontaneous at any temperature.

Practice Question 12 NEET UG — Numerical MCQ

The equilibrium constant (K) for a reaction at $300\,\mathrm{K}$ is 10. What is the value of the standard Gibbs energy change ($\Delta G^{\ominus}$) for the reaction? (Given: $R = 8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$).

(1) $-5.74\,\mathrm{kJ\,mol^{-1}}$

(2) $+5.74\,\mathrm{kJ\,mol^{-1}}$

(3) $-2.49\,\mathrm{kJ\,mol^{-1}}$

(4) $-57.4\,\mathrm{kJ\,mol^{-1}}$

Correct Answer: (1)

Worked Solution:
Using the relationship between $\Delta G^{\ominus}$ and equilibrium constant K:

$$\Delta G^{\ominus} = -2.303\,RT\log_{10} K = -2.303 \times (8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}) \times (300\,\mathrm{K}) \times \log_{10}(10)$$

$$\Delta G^{\ominus} = -2.303 \times 8.314 \times 300 \times 1 = -5744.1\,\mathrm{J\,mol^{-1}} \approx -5.74\,\mathrm{kJ\,mol^{-1}}$$

BOSONANCE

Practice Question 13

NEET UG — Assertion-Reason

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): During the isothermal free expansion of an ideal gas into vacuum, $\Delta U = 0$, $q = 0$, and $w = 0$, yet the process is spontaneous and irreversible.
Reason (R): During isothermal free expansion of an ideal gas, $\Delta S_{\text{sys}} > 0$ while $\Delta S_{\text{surr}} = 0$, making $\Delta S_{\text{total}} > 0$.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

Correct Answer: (1)

Worked Solution:
In an isothermal free expansion of an ideal gas into vacuum ($p_{\text{ex}} = 0$), no work is done ($w = 0$), and by Joule's experiment no heat is exchanged ($q = 0$), so $\Delta U = 0$. Because no heat is exchanged with the surroundings, $\Delta S_{\text{surr}} = 0$. However, entropy is a state function, so $\Delta S_{\text{sys}} = \dfrac{q_{\text{rev}}}{T} = nR\ln\left(\dfrac{V_f}{V_i}\right) > 0$. Thus $\Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$, which drives the spontaneous, irreversible expansion. Both (A) and (R) are true and (R) correctly explains (A).

Practice Question 14

NEET UG — Multi-Statement Evaluation

Consider the following two thermochemical equations at $298\,\text{K}$:
Reaction I: $\frac{1}{2}\text{N}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{NO}(\text{g}); \quad \Delta_r H^\ominus = +90\,\text{kJ mol}^{-1}$
Reaction II: $\text{NO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{NO}_2(\text{g}); \quad \Delta_r H^\ominus = -74\,\text{kJ mol}^{-1}$
Identify the correct statement(s) regarding the thermodynamic stability of $\text{NO}(\text{g})$ and $\text{NO}_2(\text{g})$:

(1) $\text{NO}(\text{g})$ is thermodynamically stable with respect to $\text{N}_2(\text{g})$ and $\text{O}_2(\text{g})$ because it absorbs heat during formation.

(2) $\text{NO}(\text{g})$ is thermodynamically unstable (endothermic formation, $\Delta_f H^\ominus = +90\,\text{kJ mol}^{-1}$) and readily oxidizes to $\text{NO}_2(\text{g})$ in an exothermic process ($\Delta_r H^\ominus = -74\,\text{kJ mol}^{-1}$).

(3) The standard enthalpy of formation ($\Delta_f H^\ominus$) of $\text{NO}_2(\text{g})$ from $\text{N}_2(\text{g})$ and $\text{O}_2(\text{g})$ is $-164\,\text{kJ mol}^{-1}$.

(4) Both Reactions I and II have $\Delta n_g = 0$.

Correct Answer: (2)

Worked Solution:
• Reaction I is strongly endothermic ($\Delta_f H^\ominus[\text{NO}(\text{g})] = +90\,\text{kJ mol}^{-1} > 0$), meaning $\text{NO}(\text{g})$ has higher enthalpy than its constituent elements and is thermodynamically unstable.
• Reaction II is exothermic ($\Delta_r H^\ominus = -74\,\text{kJ mol}^{-1} < 0$), so unstable $\text{NO}(\text{g})$ readily combines with $\text{O}_2(\text{g})$ to form $\text{NO}_2(\text{g})$.
• Adding Reactions I and II gives $\frac{1}{2}\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{NO}_2(\text{g})$ with $\Delta_f H^\ominus[\text{NO}_2(\text{g})] = +90 + (-74) = +16\,\text{kJ mol}^{-1}$ (so statement 3 is false).
• For Reaction II, $\Delta n_g = 1 - 1.5 = -0.5 \neq 0$ (so statement 4 is false).

BOSONANCE

Practice Question 15 NEET UG — Matrix Match & Calculation

When $1.00\,\text{mol}$ of liquid water, $\text{H}_2\text{O}(\text{l})$, is formed from $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$ under standard conditions at $298\,\text{K}$, $\Delta_f H^\ominus = -286\,\text{kJ mol}^{-1}$. What is the entropy change of the surroundings (ΔS_{surr}), and how does the Third Law of Thermodynamics apply to pure crystalline ice vs. supercooled liquid water at $0\,\text{K}$?

(1) $\Delta S_{\text{surr}} = +959.7\,\text{J K}^{-1}\,\text{mol}^{-1}$; entropy at $0\,\text{K}$ is zero for pure crystalline ice but non-zero for supercooled water.

(2) $\Delta S_{\text{surr}} = -959.7\,\text{J K}^{-1}\,\text{mol}^{-1}$; entropy at $0\,\text{K}$ is zero for both pure crystalline ice and supercooled water.

(3) $\Delta S_{\text{surr}} = +286.0\,\text{J K}^{-1}\,\text{mol}^{-1}$; entropy at $0\,\text{K}$ is non-zero for pure crystalline ice.

(4) $\Delta S_{\text{surr}} = +959.7\,\text{J K}^{-1}\,\text{mol}^{-1}$; entropy at $0\,\text{K}$ is zero for solutions and supercooled liquids alike.

Correct Answer: (1)

Worked Solution:

• Heat evolved by the system is absorbed by the surroundings at constant pressure and $T = 298\,\text{K}$:

$$q_{\text{surr}} = -\Delta_f H^\ominus = +286\,\text{kJ mol}^{-1} = +286 \times 10^3\,\text{J mol}^{-1}$$

$$\Delta S_{\text{surr}} = \frac{q_{\text{surr}}}{T} = \frac{+286 \times 10^3\,\text{J mol}^{-1}}{298\,\text{K}} = +959.73\,\text{J K}^{-1}\,\text{mol}^{-1}$$

• According to the Third Law of Thermodynamics, the entropy of any **pure crystalline solid** approaches zero at $0\,\text{K}$ due to perfect crystal order, whereas solutions and supercooled liquids retain positional disorder and have non-zero entropy at $0\,\text{K}$.

10. Common Pitfalls & Exam Revision Checklist

Common Pitfalls to Avoid in NEET UG

1. **Physics vs. Chemistry Sign Convention for Work:** In IUPAC chemical thermodynamics, work done **ON** the system is positive ($w > 0$) and work done **BY** the system is negative ($w < 0$), with $\Delta U = q + w$. Never mix this up with the older convention $\Delta U = q - w$.
2. **Unit Mismatch in $\Delta H = \Delta U + \Delta n_g RT$ and $\Delta G = \Delta H - T\Delta S$:** ΔH and ΔU are almost always given in kJ mol^{-1}, whereas $R = 8.314\,\text{J K}^{-1}\,\text{mol}^{-1}$ and ΔS are given in $\text{J K}^{-1}\,\text{mol}^{-1}$. Always convert R and ΔS to kJ by multiplying by 10^{-3} before adding or subtracting!
3. **Counting Solids or Liquids in Δn_g:** Only **gaseous** reactants and products are included when computing $\Delta n_g = n_g(\text{products}) - n_g(\text{reactants})$. Ignore all solids, liquids, and aqueous species.
4. **Reactants vs. Products Order in Enthalpy Formulas:**
   - Using Enthalpies of Formation ($\Delta_f H^\ominus$): $\Delta_r H^\ominus = \sum \Delta_f H^\ominus(\textbf{products}) - \sum \Delta_f H^\ominus(\textbf{reactants})$.
   - Using Bond Enthalpies ($\Delta_{\text{bond}} H^\ominus$) or Combustion Enthalpies ($\Delta_c H^\ominus$): $\Delta_r H^\ominus = \sum(\textbf{reactants}) - \sum(\textbf{products})$.
5. **Confusing ΔU and ΔS in Isothermal Expansion of an Ideal Gas:** For an ideal gas at constant T, $\Delta U = 0$ and $\Delta H = 0$ whether the expansion is reversible or irreversible. However, $\Delta S_{\text{total}} = 0$ for reversible expansion and $\Delta S_{\text{total}} > 0$ for irreversible expansion.

Concise Master Formula Recap

- **First Law of Thermodynamics:** $\Delta U = q + w$ (For isolated system: $q = 0, w = 0 \implies \Delta U = 0$)
- **Irreversible pV-Work:** $w_{\text{irrev}} = -p_{\text{ex}}\Delta V = -p_{\text{ex}}(V_f - V_i)$
- **Reversible Isothermal pV-Work (Ideal Gas):** $w_{\text{rev}} = -nRT\ln\left(\frac{V_f}{V_i}\right) = -2.303\,nRT\log_{10}\left(\frac{V_f}{V_i}\right)$
- **Enthalpy & Internal Energy Relation:** $H = U + pV \implies \Delta H = \Delta U + p\Delta V = \Delta U + \Delta n_g RT$
- **Heat Capacity Relations:** $q = C\Delta T = nC_{\text{m}}\Delta T = mc\Delta T$; $q_V = \Delta U = nC_{V,\text{m}}\Delta T$; $q_p = \Delta H = nC_{p,\text{m}}\Delta T$; $C_p - C_V = R$ (for 1 mol ideal gas)
- **Enthalpy of Solution:** $\Delta_{\text{sol}} H^\ominus = \Delta_{\text{lattice}} H^\ominus + \Delta_{\text{hyd}} H^\ominus$
- **Entropy & Second Law:** $\Delta S_{\text{sys}} = \frac{q_{\text{rev}}}{T}$; $\Delta S_{\text{surr}} = -\frac{\Delta H_{\text{sys}}}{T}$ (at constant p, T); $\Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$ (spontaneous)
- **Gibbs Energy & Equilibrium:** $\Delta G = \Delta H - T\Delta S$; $\Delta_r G^\ominus = -RT\ln K = -2.303\,RT\log_{10} K$

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